

Diagnostic Expectations and Sovereign Default*

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Abstract

The workhorse model of sovereign default prices debt as a function of current output and the stock of debt: conditional on the state, pricing has no memory. I replace the lenders' rational expectations with diagnostic expectations, a disciplined model of overreaction to news, in an otherwise standard Eaton–Gersovitz economy. One new state variable appears: the direction of recent income news. The same debt and the same output are priced differently after good news than after bad, and expected bond returns are predictably low after booms. Default regions expand when optimism reverses. The overreaction parameter is measured from forecast-revision regressions, not fitted to spreads, and the mapping from the Coibion–Gorodnichenko coefficient to the model is exact. At the survey-consistent parameter, a boom-reversal quarter carries a default hazard of 12.6 percent against 1.5 percent under rational pricing on the identical income path, a gap that survives recalibrating the non-belief parameters to restore the observed default frequency. Whether fiscal rules are worth having turns on whose beliefs are distorted: a debt ceiling lowers sovereign welfare when only the market overreacts, and raises it when the government shares the market's beliefs.

JEL codes: E44, F34, G41, H63.

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1 Introduction

In June 2017 Argentina sold \$2.75 billion of a hundred-year dollar bond priced to yield 7.9 percent, and investors placed orders for roughly three times that amount. Argentina is a serial defaulter by any standard: Reinhart and Rogoff (2009) count seven external defaults or restructurings since independence, and the 2001 default was then the largest in history. Less than three years after the century bond was sold, in May 2020, the country was in default for the ninth time. A creditor who bought the bond at issue and held it to the restructuring lost most of his investment.

What should we make of episodes like this one? One reading is that fundamentals moved: commodity prices and fiscal outcomes both deteriorated between 2017 and 2020, and so did

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politics. They did. But the pricing of a hundred-year claim on a serial defaulter should not swing from enthusiasm to panic on three years of ordinary macroeconomic news, and the Argentine episode is not an isolated case. Emerging-market lending arrives in waves. Spreads compress during booms and blow out afterward, and measures of credit-market sentiment forecast low or negative returns to creditors, in sovereign markets as in corporate ones (Greenwood and Hanson, 2013; López-Salido et al., 2017). Reinhart and Rogoff (2009) give the pattern a name: “this time is different.” The reading I pursue in this paper is that beliefs move more than fundamentals do, in a particular, measurable way, and that this movement changes what sovereign debt models predict and what fiscal policy should do.

The workhorse theory of sovereign borrowing, from Eaton and Gersovitz (1981) through its quantitative implementations by Aguiar and Gopinath (2006), Arellano (2008), and Chatterjee and Eyigungor (2012), explains spreads through the option to default under incomplete markets. The theory has real successes: default is chosen in bad times, and the countercyclicality of spreads comes out right along with the procyclicality of borrowing. It also has a striking property. Every equilibrium object, the price schedule as much as the default set, is a function of the current state, debt b and income y . Two countries with identical debt and identical income borrow on identical terms, whatever histories brought them to that point. Conditional on the state, pricing has no memory.

This paper changes one primitive of that model and nothing else: the process by which creditors form beliefs. Creditors hold *diagnostic expectations* in the sense of Bordalo et al. (2018): after observing news, they overweight future states whose likelihood the news has raised. The distortion has a kernel of truth, since beliefs respond to genuine information and in the right direction. The failure is one of degree, governed by a single parameter $\theta \geq 0$, with rational expectations recovered exactly at $\theta = 0$. When log income follows a Gaussian autoregression, the diagnostic conditional distribution is Gaussian with the same variance and a mean shifted by $\theta\rho(x - h)$, where x is current log income and h is last period’s (Lemma 1). Creditors who have just seen income rise expect the improvement to continue, and creditors who have just seen it fall expect further decline.

The consequence is that one new state variable enters the pricing of sovereign debt: the direction of recent news, $\nu = x - h$. I put diagnostic creditors inside an otherwise standard one-period-bond default economy and establish three results. First, the equilibrium price schedule is history dependent: at fixed debt and fixed income, bond prices increase in ν , strictly so wherever default risk is interior (Proposition 1). The rational model predicts a zero coefficient on news once debt and income are controlled for, and the diagnostic model predicts a positive one. That zero is the natural test of the theory. Second, expected bond returns under the true income process are predictably low after good news and predictably high after bad news (Proposition 2), which is the sovereign analogue of the return predictability that credit-market sentiment generates in corporate bonds. Third, default regions are history dependent: debt that is safe after good news becomes default-inducing when the news reverses, so the model produces boom-bust credit cycles from ordinary shocks, with no disasters and no self-fulfilling runs (Proposition 3). Each result is

a comparative static conditional on one regularity condition, monotone default in income, whose standing I discuss and verify at length below.

The discipline of the exercise is that θ is not free. Diagnostic expectations imply overreaction in individual forecasts: a forecaster who revises upward tends to be disappointed, so realized forecast errors are negatively related to forecast revisions, which is precisely what Bordalo et al. (2020) document across United States macroeconomic series in the framework of Coibion and Gorodnichenko (2015). In the model economy the mapping between the forecast-error-on-revision coefficient β_{CG} and θ is exact (Proposition 4): $\beta_{CG} = -\theta(1 + \theta)/[(1 + \theta)^2 + \theta^2\rho^2]$. Published estimates of overreaction put θ between roughly 0.5 and 1 (Bordalo et al., 2018; Bordalo et al., 2020). I take $\theta = 0.5$, which corresponds to $\beta_{CG} = -0.30$, as the baseline, and I specify the forecast-revision regressions for emerging-market growth forecasts that would estimate θ country by country.

Quantitatively, the belief distortion does a specific kind of work. I solve the model on the Arellano (2008) calibration for Argentina, first changing only θ as a controlled comparative static, and then re-selecting the discount factor and the default cost with θ held at its survey value, since a mechanism that changes default incentives invalidates a calibration built without it. At $\theta = 0.5$, a one-standard-deviation piece of good news lowers the spread by 462 basis points holding debt and output fixed, where the rational benchmark says the effect should be zero. The standard deviation of spreads rises by a factor of 8.6 while the median spread *falls* from 3.6 to 1.6 percent: long calm spells punctuated by crises. The sharpest difference is dynamic. On the identical simulated income path, I locate every episode in which income rises for three or more consecutive quarters and then falls. In the quarter of the reversal the default hazard is 12.6 percent in the diagnostic economy and 1.5 percent in the rational one. Same fundamentals, same shocks: the difference is entirely the pricing of beliefs, and it is the model’s account of why defaults so often follow booms. The recalibrated economy tells the same story in more moderate numbers, a reversal-quarter hazard of 3.1 percent against 0.3, and I report both throughout.

The policy question is then whether governments should bind themselves against belief-driven cycles, and here the model delivers an answer with an edge to it. Welfare depends on *whose* expectations are diagnostic. If the government is rational while the market overreacts, debt limits make the sovereign worse off: the government is selling overpriced claims to optimists and repurchasing them cheap from pessimists, and a ceiling interferes with a trade that benefits it. If the government shares the market’s beliefs, which is what survey evidence on official and professional forecasters suggests, the ranking reverses: a ceiling that binds after good-news sequences prevents borrowing that the government would not choose under the true distribution, and the welfare gain is increasing in θ . I compute the full two-by-two of government and market beliefs. The lesson for the fiscal-rules debate is that rules are a commitment device against the government’s own extrapolation, not a correction of market mispricing.

The paper joins two literatures and takes one step beyond each. The quantitative sovereign default literature (Aguiar and Gopinath, 2006; Arellano, 2008; Chatterjee and Eyigungor, 2012; Mendoza and Yue, 2012; Hatchondo and Martinez, 2009), surveyed in Aguiar et al. (2016),

prices debt under rational expectations, and belief-driven crises enter it through equilibrium multiplicity, as in Cole and Kehoe (2000) and Lorenzoni and Werning (2019). Multiplicity and overreaction are different theories with different footprints. A sunspot changes outcomes with no news about fundamentals. Diagnostic beliefs distort the response to genuine news in a fixed direction and by a measurable amount, and forecast data discipline that amount rather than leaving it to the modeler. The behavioral literature (Bordalo et al., 2018; Gennaioli and Shleifer, 2018; Bordalo et al., 2020; Bordalo et al., 2022) has established overreaction in forecasts and linked credit-market sentiment to reversals (Greenwood and Hanson, 2013; López-Salido et al., 2017), but it has not, to my knowledge, been put inside the default option that gives sovereign spreads their nonlinearity. The marriage is what produces the new content: history-dependent default regions and the whose-beliefs answer to the fiscal-rules question, neither of which arises in a linear credit model or a rational default model.

My plan is as follows. Section 2 presents a two-period example that contains the paper’s main mechanisms in closed form. Section 3 sets out the full model and defines equilibrium. Section 4 states the theoretical results. Section 5 maps survey overreaction to θ and specifies the measurement design. Section 6 reports the quantitative results, Section 7 the spread-panel evidence and its design, and Section 8 the fiscal-rule experiments. Section 9 concludes. Proofs and extensions are in the appendix.

2 A Two-Period Example

Nearly everything in this paper can be seen in a two-period economy, so I begin there. A sovereign has endowment y_0 at date 0 and a random endowment at date 1 that takes the value y_H with true probability π and $y_L < y_H$ otherwise. At date 0 it sells one-period bonds with face value b to competitive risk-neutral foreign lenders whose opportunity cost is the gross rate R . At date 1 it repays or defaults. Default costs a fraction ψ of that period’s endowment, so the sovereign repays exactly when $b \leq \psi y_1$. I restrict attention to the interesting region $\psi y_L < b \leq \psi y_H$, where debt is repaid in the high state and repudiated in the low state.

Lenders do not use π . At date 0 they have just observed news about the country: formally, the current probability π differs from a reference probability π_0 that summarizes beliefs before the news arrived. Diagnostic lenders overweight the states the news has made more representative. I take the linear form

$$\hat{\pi} = \pi + \theta(\pi - \pi_0), \quad \theta \geq 0, \quad (1)$$

which is the two-state counterpart of the density tilt introduced in Section 3. The news is $\nu = \pi - \pi_0$, and $\theta = 0$ recovers rational expectations. Competitive risk-neutral pricing gives the bond price

$$q = \frac{\hat{\pi}}{R} = \frac{\pi + \theta\nu}{R}. \quad (2)$$

Equation (2) already contains the paper’s first result. Fix fundamentals: two countries with the same current repayment probability π , the same debt, the same output. If one arrived at

π from below (good news, $\nu > 0$) and the other from above, the first borrows more cheaply. History matters for prices even though it carries no information about repayment, because history determines what the news was, and news is what diagnostic beliefs overweight.

The second result concerns returns. A lender who buys at price q receives 1 with true probability π , so his expected gross return is $\pi/q = R\pi/\hat{\pi}$, and the expected excess return is

$$\mathbb{E}[R^e] = R \frac{\pi - \hat{\pi}}{\hat{\pi}} = - \frac{R\theta\nu}{\pi + \theta\nu}. \quad (3)$$

After good news, creditors systematically overpay and earn low returns. After bad news the reverse holds. At $\theta = 0$ the excess return is zero for every ν : predictable returns from public news are exactly the rational model's forbidden pattern, which is what makes them evidence.

The third result concerns quantities. Let the sovereign have log utility, discount factor β , and rational beliefs π . Only the market is diagnostic for now. It chooses b to maximize $\ln(y_0 + qb) + \beta[\pi \ln(y_H - b) + (1 - \pi) \ln((1 - \psi)y_L)]$. The first-order condition gives, after rearrangement,

$$b^* = \frac{y_H}{1 + \beta\pi} - \frac{\beta\pi R y_0}{\hat{\pi}(1 + \beta\pi)}, \quad (4)$$

which is strictly increasing in $\hat{\pi}$ and hence in the news ν . Optimism it does not share still moves the sovereign: a higher $\hat{\pi}$ raises the price at which it can sell claims on the high state, and it rationally sells more of them. Borrowing responds to market sentiment about fundamentals over and above fundamentals themselves. There is also a margin in ν at which the sovereign switches from safe debt ($b \leq \psi y_L$) to risky debt, since the value of the risky plan rises with $\hat{\pi}$ while the safe plan's value does not. Optimism moves the economy from a region where default never happens to one where it happens with probability $1 - \pi$. Good news, filtered through diagnostic pricing, buys fragility.

Whose welfare does the distortion damage? Here the example teaches a lesson that survives, quantitatively, in the full model. The rational sovereign in (4) is exploiting mispricing, not suffering from it: it sells overpriced claims after good news, and the expected losses in (3) fall on the lenders. Capping b would deny the sovereign a profitable trade. The case for a borrowing limit appears only when the government itself prices the future with $\hat{\pi}$ rather than π , so that its own plans embody the extrapolation. Then a rule chosen behind the veil, before the news arrives, can raise the government's welfare evaluated under the truth. Section 8 makes this precise and puts numbers on it.

What the example cannot address is dynamics: the two-period structure has no room for the history dependence of *default sets* or for booms that build over several quarters, and it is silent on how belief distortions interact with the equilibrium recursion that maps future default behavior into today's prices. For that I need the full model, to which I now turn.

3 The Model

3.1 Environment

Time is discrete and infinite. A small open economy receives a stochastic endowment y_t whose logarithm $x_t = \log y_t$ follows

$$x_{t+1} = \rho x_t + \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim N(0, \sigma_\varepsilon^2), \quad 0 < \rho < 1. \quad (5)$$

The government is benevolent, with preferences $\mathbb{E}_0 \sum_t \beta^t u(c_t)$ and $u(c) = c^{1-\gamma}/(1-\gamma)$. It borrows by selling one-period non-contingent bonds to competitive, risk-neutral foreign lenders whose outside option is the risk-free gross rate $R = 1 + r^*$. A bond with face value b' sold today is a promise to pay b' tomorrow. The government cannot commit to honor it.

At the start of a period the government owes $b \geq 0$ and chooses whether to repay or default. Repayment means paying b and selling new debt $b' \in \mathcal{B} \subset \mathbb{R}_+$ at the market price q , so that consumption is $c = y + qb' - b$. Default means the debt is written off in full, but the country is excluded from markets and suffers the output loss standard in this literature: income during exclusion is $y^D(x) = \min\{y, \phi \mathbb{E}[y]\}$, an asymmetric cost that bites in good states, which Arellano (2008) shows is needed for default to be chosen in equilibrium at realistic frequencies and which Mendoza and Yue (2012) derive from imported-input substitution. Exclusion ends with probability λ each period, at which point the country re-enters with zero debt.

All of this is the workhorse model of Eaton and Gersovitz (1981) and Arellano (2008). The single departure is next.

3.2 Creditor beliefs

Lenders form beliefs about x' by distorting the true conditional density $p(x' | x)$ toward states that recent news has made more representative. Let $h = x_{-1}$ denote last period's log income. Following Bordalo et al. (2018), the diagnostic density is

$$\hat{p}_\theta(x' | x, h) = p(x' | x) \left[\frac{p(x' | x)}{p(x' | h)} \right]^\theta \frac{1}{Z(x, h)}, \quad \theta \geq 0, \quad (6)$$

where Z is the normalizing constant. The reference density $p(\cdot | h)$ is the forecast an analyst would have made had the economy stood still: the distortion overweights precisely those states whose odds have improved relative to that reference, which is the “kernel of truth” property. At $\theta = 0$ beliefs are rational. The distortion is not pessimism or optimism as a trait. It is an amplifier of whatever the news was.

For the Gaussian autoregression (5) the tilt takes a form that makes the model tractable.

Lemma 1 (Diagnostic tilt). *Let $p(\cdot | x)$ be the $N(\rho x, \sigma_\varepsilon^2)$ density and $p(\cdot | h)$ the $N(\rho h, \sigma_\varepsilon^2)$ density. Then $\hat{p}_\theta(\cdot | x, h)$ is the $N(\rho x + \theta \rho(x - h), \sigma_\varepsilon^2)$ density.*

The proof, a matter of completing a square, is in Appendix A. Three features deserve notice.

The variance is undistorted: diagnostic creditors are not more or less uncertain, they are wrong about the mean. The mean distortion is proportional to the news, $\nu \equiv x - h$: after a quarter in which log income rose, creditors expect the rise to continue beyond what persistence justifies. And the belief state is just the pair (x, h) , so the entire departure from rational expectations costs one state variable, and that variable is observable in the data as a growth realization or a forecast revision.

Two modeling choices require defense. First, why does competition not eliminate the distortion? The lenders in this model are the marginal pricers of emerging-market debt, and the overreaction I attribute to them is documented in the forecasts of exactly this population: professional analysts, individually, revise too far in the direction of news (Bordalo et al., 2020; Bordalo et al., 2022). Arbitraging a belief distortion in a defaultable bond requires taking large positions against it and bearing the default risk meanwhile, and the capital that does this is limited, as the long literature on credit-market sentiment attests (Greenwood and Hanson, 2013; López-Salido et al., 2017). I therefore let the price be set each period by risk-neutral diagnostic lenders, the exact behavioral counterpart of the rational risk-neutral pricing in the workhorse model. Second, whose beliefs are diagnostic? In the baseline the government is rational and only the market overreacts, so that every pricing result isolates the market's distortion. The government's own beliefs matter for normative questions, and Section 8 treats the full two-by-two.

3.3 Recursive formulation and equilibrium

The aggregate state for pricing is (b', x, h) . The government's value satisfies

$$V(b, x, h) = \max \left\{ V^R(b, x, h), V^D(x) \right\}, \quad (7)$$

with repayment value

$$V^R(b, x, h) = \max_{b' \in \mathcal{B}} \left\{ u(y + q(b', x, h) b' - b) + \beta \mathbb{E}[V(b', x', x) | x] \right\}, \quad (8)$$

where the expectation is under the true kernel (5) because the government is rational, and default value

$$V^D(x) = u(y^D(x)) + \beta \mathbb{E}[\lambda V(0, x', x) + (1 - \lambda)V^D(x') | x]. \quad (9)$$

Note the bookkeeping in the continuation values: tomorrow's reference point is today's income, $h' = x$. Let $d(b, x, h) \in \{0, 1\}$ indicate default. Lenders break even under *their* beliefs:

$$q(b', x, h) = \frac{1}{R} \widehat{\mathbb{E}}_\theta[1 - d(b', x', x) | x, h], \quad (10)$$

where $\widehat{\mathbb{E}}_\theta$ integrates against $\hat{p}_\theta(\cdot | x, h)$. Equation (10) is where history enters: h appears in the price because h shapes the belief, even though h carries no information about x' under the truth.

Definition 1. A recursive equilibrium with diagnostic pricing is a value function V , policies d and $b'(\cdot)$, and a price schedule q such that (i) given q , the value and policies solve (7)–(9), and

(ii) given d , the price satisfies (10) for all (b', x, h) .

This is a temporary-equilibrium concept: agents optimize given stated beliefs and markets clear, while the beliefs of one side are systematically wrong under the true measure. Existence follows from the same arguments as in the rational case, since $\hat{p}_\theta$ is a proper Markov kernel on the expanded state (x, h) and the price operator maps the compact set $[0, 1/R]^{|\mathcal{B}| \times |\mathcal{X}|^2}$ into itself. I compute the equilibrium as the limit of the standard iteration, which is the selection used throughout the quantitative literature (Aguiar et al., 2016). At $\theta = 0$, h drops out of (10) and the model is exactly Arellano (2008).

The question the theory must answer is what the extra state variable does to prices, returns, and default sets. That is the next section.

4 Theoretical Results

Throughout, $\nu = x - h$ denotes the news and $F_\theta(\cdot \mid x, h)$ the diagnostic conditional cdf. Because the referee-grade distinction between what is proved and what is computed matters for everything that follows, I state the epistemic status of each result where it appears. Lemmas 1 and 2 and Proposition 4 are exact properties of the belief kernel and involve no equilibrium objects. Lemma 3 proves the debt margin of monotone default in any equilibrium. Propositions 1 to 3 are comparative statics conditional on Assumption 1, the income margin of monotone default, which resists proof at this level of generality but is verified without exception in every computed equilibrium, across grid densities, default-cost specifications, and parameter vectors (Appendix C). The borrowing response to news is a numerical property of the computed equilibrium, and I flag it as such when it arrives. The welfare magnitudes of Section 8 are numerical properties of stated parameterizations.

Lemma 2 (News orders beliefs). *Fix x . If $\nu_2 > \nu_1$ then $F_\theta(\cdot \mid x, h_2)$ first-order stochastically dominates $F_\theta(\cdot \mid x, h_1)$, where $h_i = x - \nu_i$, with strict dominance whenever $\theta > 0$.*

Lemma 3 (Default is monotone in debt). *In any equilibrium of Definition 1, for every (x, h) : the repayment value $V^R(b, x, h)$ is nonincreasing in b , strictly decreasing wherever finite, $V^D(x)$ does not depend on b , and the default set $\{b \in \mathcal{B} : d(b, x, h) = 1\}$ is an upper interval $\{b \geq \underline{b}(x, h)\}$.*

Lemma 3 needs no regularity conditions: a larger inherited debt lowers consumption at every issuance choice and leaves the continuation problem unchanged, while the default value never involves b . The proof is three lines in Appendix A. What cannot be proved at this level of generality is the income margin, which I now isolate as the paper's single maintained assumption.

Assumption 1 (Monotone default in income). *The default policy $d(b, x, h)$ is nonincreasing in x for fixed (b, h) .*

Assumption 1 says default is a bad-times event. Arellano (2008) proves it for i.i.d. income. With persistent income it is the maintained regularity condition of the quantitative literature

even at $\theta = 0$, because repayment and default values both rise with income and their ranking is not globally orderable by primitive conditions alone. My response is systematic verification rather than assertion: the condition holds at every grid point of every computed equilibrium at $\theta \in \{0, 0.5\}$, it continues to hold when both grids are doubled ($n_x = 31$, $n_b = 400$), under a proportional default cost in place of the asymmetric one, and at every parameter vector visited by the recalibration search of Section 6 (Appendix C reports the counts, and the only violations anywhere are 0.1 percent of columns at $\theta = 1$, traceable to near-degenerate pricing at extreme states). A theorem covering the income margin under primitive conditions would be better, and I do not have one. Neither does the rational literature.

Proposition 1 (History-dependent prices). *Under Assumption 1, the price schedule $q(b', x, h)$ is nondecreasing in ν for every (b', x) , and strictly increasing in ν at every (b', x) where $\theta > 0$ and next-period default risk is interior.*

Corollary 1 (The rational zero). *Conditional on (b', x) , prices and spreads do not vary with ν if and only if $\theta = 0$ (given interior default risk somewhere). In particular, in any panel regression of spreads on news within cells defined by debt and output, the rational model predicts a coefficient of exactly zero.*

Proposition 1 is the paper’s pricing result and Corollary 1 its identification content. Debt and output are controls and the news is the treatment. The workhorse model supplies the sharp null. I emphasize what the corollary does not say: it does not say spreads are unrelated to growth, which they are under both hypotheses through x itself. The test lives entirely inside the conditional.

Proposition 2 (Predictable returns). *Let $R^e(b', x, h)$ denote the true-measure expected gross return on the bond in excess of R . Under Assumption 1, R^e is nonincreasing in ν , with $R^e = 0$ for all ν when $\theta = 0$, $R^e \leq 0$ for $\nu > 0$, and $R^e \geq 0$ for $\nu < 0$ when $\theta > 0$ (strictly where default risk is interior).*

The mechanism is the one in equation (3): after good news the lenders’ repayment probability exceeds the true one, so they overpay. Proposition 2 turns the sentiment-reversal evidence from corporate credit (Greenwood and Hanson, 2013; López-Salido et al., 2017) into a prediction for sovereign bonds, with the news direction as the conditioning variable.

Proposition 3 (Bust fragility). *Under Assumption 1, the repayment value $V^R(b, x, h)$ is nondecreasing in ν and $V^D(x)$ is independent of ν . Hence the default set $D(x, h) = \{b : d(b, x, h) = 1\}$ is weakly decreasing in ν : any debt level that triggers default after good news triggers it after bad news, and not conversely.*

Proposition 3 is where the model earns the word *cycle*. A sovereign that accumulated debt while $\nu > 0$, at prices that made the debt look cheap, faces a strictly larger default region the moment ν turns negative, at the same debt and the same income level. Nothing self-fulfilling is involved: the price schedule is a function, not a correspondence, and there are no sunspots to

coordinate on, in contrast to Cole and Kehoe (2000) and Lorenzoni and Werning (2019). The fragility is manufactured by the boom itself.

Borrowing behavior along the boom completes the account, and here I am deliberately more modest. Whether issuance rises with ν involves an income effect that resists a general theorem: better prices raise current consumption, which lowers the marginal utility that borrowing serves. In the two-period example the closed form (4) settles the question, and in the computed equilibria of Section 6 issuance is increasing in ν throughout the relevant region. I report it as a quantitative property, not a theorem.

These results say what diagnostic pricing does. Whether it does so *in fact* depends on a single number, θ , and the model had better not be allowed to choose it freely. The next section takes the measurement of θ out of the spread data entirely.

5 Measuring Overreaction

5.1 Forecast errors and revisions

Diagnostic expectations leave fingerprints in forecast data. Consider a forecaster predicting x_{t+1} under the beliefs of Section 3.2, and define the forecast revision $\text{Rev}_t = \hat{\mathbb{E}}_t[x_{t+1}] - \hat{\mathbb{E}}_{t-1}[x_{t+1}]$ and the forecast error $\text{FE}_{t+1} = x_{t+1} - \hat{\mathbb{E}}_t[x_{t+1}]$. Coibion and Gorodnichenko (2015) proposed regressing errors on revisions:

$$\text{FE}_{t+1} = \alpha + \beta_{CG} \text{Rev}_t + u_{t+1}. \quad (11)$$

Under rational expectations $\beta_{CG} = 0$: revisions are part of the information set. Under information rigidity, consensus forecasts underreact and $\beta_{CG} > 0$, which is what Coibion and Gorodnichenko (2015) find for consensus data. Under diagnostic expectations, individual forecasts overreact: an upward revision predicts disappointment, $\beta_{CG} < 0$, which is what Bordalo et al. (2020) find for individual professional forecasters across most United States macroeconomic series. The two findings coexist because averaging across forecasters with dispersed information pushes the consensus coefficient up. The individual-level coefficient is the one that identifies θ , and individual-level data are therefore essential: a consensus-based estimate would confound overreaction with the mechanical underreaction of averages and could even reverse its sign.

In the model economy the mapping from β_{CG} to θ is exact.

Proposition 4 (Measurement mapping). *Let x_t follow (5) and let one-period-ahead forecasts be diagnostic with parameter θ , with revisions defined on the fixed target x_{t+1} . Then the population coefficient in (11) is*

$$\beta_{CG} = -\frac{\theta(1+\theta)}{(1+\theta)^2 + \theta^2\rho^2}. \quad (12)$$

The derivation is in Appendix A. The function is zero at $\theta = 0$ and strictly decreasing, and it is bounded below, so any empirical $\beta_{CG} \in (-1/2, 0)$ inverts to a unique θ . At the calibrated persistence $\rho = 0.945$, $\theta = 0.25$ gives $\beta_{CG} = -0.19$, $\theta = 0.5$ gives -0.30 , and $\theta = 1$ gives -0.41 .

Published individual-level estimates for output growth and related series cluster in the range that inverts to θ between one-quarter and one (Bordalo et al., 2020), and the credit-cycle estimates in Bordalo et al. (2018) put θ near 0.9. I take $\theta = 0.5$ as the baseline and report $\theta \in \{0, 1\}$ throughout. Until the emerging-market estimation of the next subsection is run, $\theta = 0.5$ has the status of a scenario value chosen inside the published United States range, and I label it so in the calibration audit of Table 5.

5.2 Design for emerging-market forecasts

The estimates just cited are for the United States. The model is about emerging markets, and the honest version of the discipline requires θ estimated from forecasts of the same economies whose bonds are being priced. The design is as follows. The estimates await data access and are marked pending.

Consensus Economics has surveyed professional forecasters, bank by bank, on annual real GDP growth for most EMBI countries monthly since the early-to-mid 1990s. The forecasts are fixed-event (current and next calendar year). I convert them to a fixed twelve-month horizon by the standard weighting of the two events by months remaining, and I measure realized growth by first-release outturns to keep data revisions out of the forecast error. The estimating equation, at the forecaster (j), country (i), month (t) level,

$$\text{FE}_{i,j,t} = \alpha_i + \tau_t + \beta_{CG} \text{Rev}_{i,j,t} + u_{i,j,t}, \quad (13)$$

with standard errors clustered by country and time, delivers a pooled β_{CG} . Running (13) country by country delivers $\theta_i = \theta(\beta_{CG,i})$ by inversion of (12), adjusting the mapping for the horizon and time aggregation as derived in Appendix E. Two econometric threats deserve flagging now. Measurement error in reported forecasts attenuates Rev coefficients toward zero, and since the same forecast enters FE and Rev with opposite signs, classical noise biases β_{CG} *negative*. The standard remedy is instrumenting the revision with its own lag or with other forecasters' revisions, and both instruments are feasible in this panel. And overlapping fixed-event horizons induce moving-average errors, which the clustering must respect. Neither threat is exotic, and both have known treatments.

The payoff of the measurement is that the quantitative model of the next section has no free belief parameter: θ comes from forecasts, everything else from Arellano (2008), and the spread implications are then predictions rather than fits.

6 Quantitative Analysis

6.1 Calibration and solution

I solve the model on the quarterly calibration of Arellano (2008) for Argentina, reproduced in Table 1 with each parameter classified as externally fixed, directly estimated, internally calibrated, or scenario. The exercise proceeds in two steps whose epistemic status differs. The first varies

Table 1: Calibration (quarterly).

Parameter		Value	Category	Source or target
Risk aversion	γ	2	externally fixed	standard value in this literature
Discount factor	β	0.953	internally calibrated	default moments at $\theta = 0$ (A08)
Risk-free rate	r^*	0.017	externally fixed	U.S. five-year bond yield (A08)
Income persistence	ρ	0.945	directly estimated	detrended Argentine real GDP (A08)
Income innovation s.d.	σ_ε	0.025	directly estimated	detrended Argentine real GDP (A08)
Re-entry probability	λ	0.282	externally fixed	market re-access delays (A08)
Default output cost	ϕ	0.969	internally calibrated	spread moments at $\theta = 0$ (A08)
Overreaction	θ	$\{0, 0.5, 1\}$	scenario (published range)	Section 5

Notes: A08 is Arellano (2008). The category column applies the four-way classification of the calibration audit (externally fixed, directly estimated, internally calibrated, scenario). Parameters marked internally calibrated at $\theta = 0$ are re-selected at $\theta = 0.5$ in Section 6.5. Table 5 in Appendix C reports units, equation locations, sensitivity ranges, and headline-result effects for every row.

only the belief parameter $\theta \in \{0, 0.5, 1\}$ at fixed non-belief parameters. This is a controlled comparative static: it asks what the measured belief distortion does to a known benchmark, and its results are properties of that benchmark, not claims of fit, because β and ϕ were selected by Arellano (2008) to hit default moments at $\theta = 0$. The second step, in Section 6.5, re-selects (β, ϕ) with θ fixed at its survey value and reports which results survive. Appendix C contains the full calibration audit (Table 5): equation location, units, source, sensitivity range, and the effect of each parameter on the headline numbers. The income process is discretized on a 21-state Tauchen (1986) chain and debt on a 200-point grid. The equilibrium is computed by iterating the value function and the price schedule jointly to a tolerance of 5×10^{-8} , and the diagnostic kernel is rediscritized exactly as a mean-shifted Gaussian on the same grid (Lemma 1). A full solution takes seconds, since the belief state adds a single grid dimension to the pricing equation and none to the government’s continuation values.

6.2 Prices and default sets

Figure 1 shows the equilibrium price schedule $q(b', x, h)$ at median income for three histories: good news, no news, and bad news, each against the rational schedule, which is a single curve because h does not enter it. The histories differ only in where income stood last quarter. Current fundamentals are identical. Diagnostic pricing fans the schedule out. After good news the sovereign can carry more debt at any price, and the maximum revenue it can raise from the market, $\max_{b'} qb'$, is larger. After bad news the schedule collapses inward, and debt levels that were financeable a quarter ago no longer are. The vertical distance between the good-news and bad-news curves at intermediate debt is the price of history, and it is large precisely in the region where default risk is interior, as Proposition 1 says it must be.

Figure 2 translates prices into default behavior: it plots the smallest debt at which the sovereign defaults, as a function of income, for the same three histories. This is Proposition 3 drawn. Between the good-news and bad-news boundaries lies what I will call the fragility corridor: combinations of debt and income that are safe if the economy arrived on an upswing

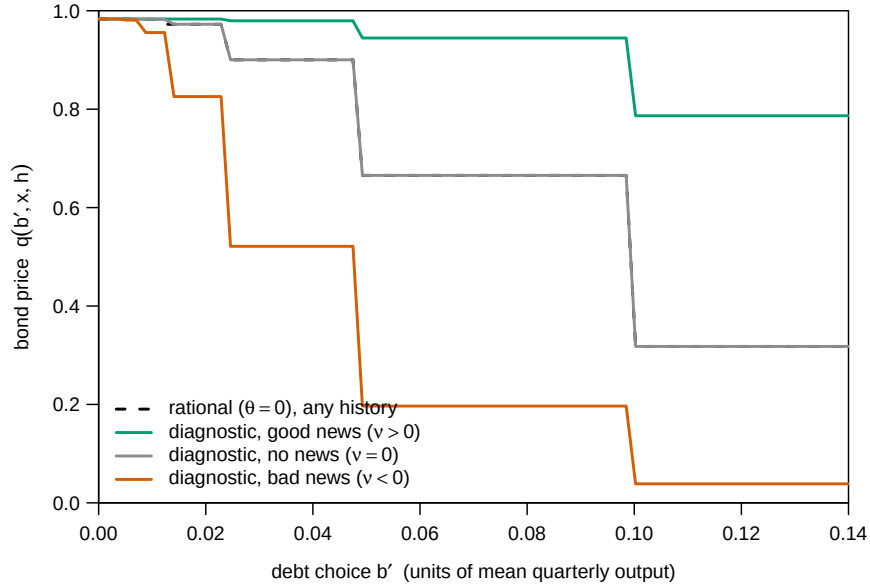


Figure 1: Bond price schedules at median income, $\theta = 0.5$, for three histories of news $\nu = x - h$ (good and bad news set $|\nu|$ to about 2.7 standard deviations of the quarterly innovation for visibility). The dashed curve is the rational schedule, common to all histories.

and default-inducing if it arrived on a downswing. A sovereign whose debt sits in the corridor is betting on the sign of the next quarter's news. The rational model has no corridor at all.

6.3 Simulated moments

Table 2 reports moments from 500,000 simulated quarters under the *true* income process, the same shock sequence for every θ .

Three rows carry the message. The news coefficient is zero to machine precision at $\theta = 0$: this is Corollary 1 holding in the simulated data, and it bears emphasis that the zero requires nonparametric conditioning. A linear regression of spreads on news with linear debt and output controls yields -168 basis points per standard deviation in the *rational* economy, an artifact of the spread schedule's curvature loading on the news term. At $\theta = 0.5$ the same linear specification even gets the sign wrong, because good news shifts issuance along the schedule. Within cells, the diagnostic economy prices news at -462 basis points per standard deviation and the rational economy at zero. The empirical work in Section 7 inherits this lesson directly: the news test must condition on debt and output flexibly, or it tests nothing.

Second, spread volatility. The workhorse model's known weakness with one-period debt is that spreads are too stable. Diagnostic pricing multiplies the standard deviation of spreads by roughly 8.6 while *lowering* the median spread from 3.6 to 1.6 percent. The economy spends most of its time calm, with compressed spreads, and occasionally explodes: long quiet spells punctuated by crises is exactly the shape of emerging-market spread data, and it emerges here from Gaussian shocks alone, with no disasters and no regime switches.

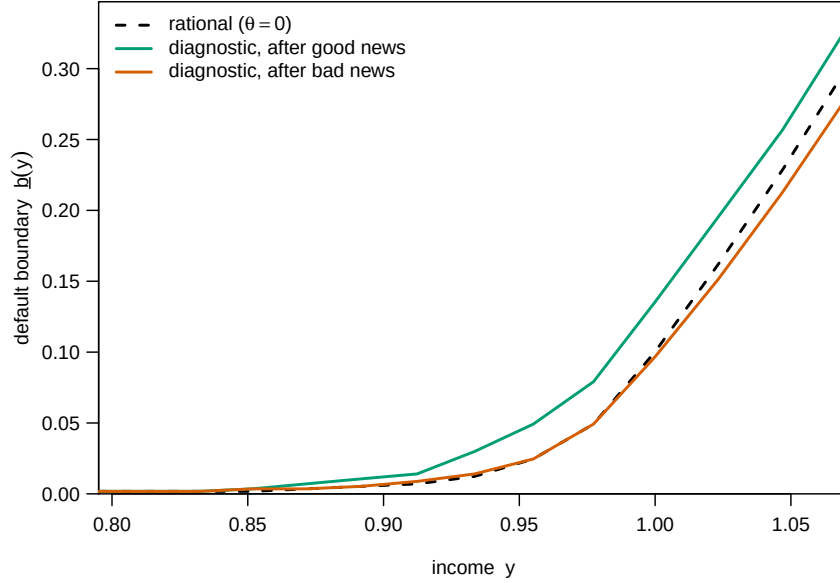


Figure 2: Default boundaries $b(y)$: the smallest debt triggering default, by income, $\theta = 0.5$. The region between the good-news and bad-news curves is history dependent.

Third, defaults. The default frequency rises from 3.7 to 11.6 per century, overshooting the 2–3 per century in Argentine data. I report this rather than hide it: the non-belief parameters were calibrated by Arellano (2008) to hit the default frequency *at* $\theta = 0$, so adding overreaction at fixed parameters must raise it. Table 2 should therefore be read as comparative statics in θ , not as the model’s best fit. Section 6.5 re-selects (β, ϕ) at the survey value of θ and reports what survives, and the history-dependence results are not at stake in that exercise, because they follow from Propositions 1–3 at any parameter vector satisfying Assumption 1. The $\theta = 1$ column makes the fixed-parameter point in extreme form: at credit-cycle levels of overreaction the economy loses market access a quarter of the time, which is a warning against importing θ estimates across asset classes without re-estimating the environment.

6.4 Booms that end

The model’s dynamic content is easiest to see in an event study, and the simulation design allows a clean one: because the income path is exogenous and the shock sequence is common across θ , I can locate every episode, 5,593 of them in half a million quarters, in which log income rises for at least three consecutive quarters and then falls, and compare the rational and diagnostic economies *on the same episodes*. Figure 3 shows median spreads, mean debt, and the default hazard in event time.

The figure shows the whole belief cycle. Entering the episode, on the heels of whatever bad news preceded the boom, the diagnostic economy pays spreads several times the rational level: pessimism is extrapolated too. The boom then compresses diagnostic spreads *below* the rational benchmark, quarter by quarter, while debt rises by 136 percent against 123 percent in

Table 2: Simulated moments under the true income process.

	$\theta = 0$	$\theta = 0.5$	$\theta = 1$
Median spread (percent, annualized)	3.6	1.6	0.5
S.d. of spread (percent)	6.8	57.8	1,277.6
Correlation of spread with log income	-0.29	-0.17	-0.28
Mean debt service to output (percent)	4.4	3.8	6.1
Defaults per 100 years	3.7	11.6	29.3
Share of quarters with market access (percent)	96.7	89.7	73.9
News coefficient, within (b', y) cells (bp per 1σ)	0.0	-462	-21,313
Implied forecast-revision coefficient β_{CG}	0	-0.30	-0.41

Notes: spreads are annualized from one-period bond prices and winsorized at the 1st and 99th percentiles before computing means and standard deviations. Medians are unaffected. The news coefficient is the slope of the winsorized spread on ν after demeaning both within cells defined by the debt choice and the income state, scaled by one standard deviation of ν . It is the model analogue of a panel regression of spreads on forecast revisions with nonparametric controls for debt and output. All statistics condition on market access.

the rational economy: the diagnostic sovereign borrows into the optimism, as the two-period example said it rationally should. The difference arrives with the reversal. In the quarter the news turns, the default hazard reaches 12.6 percent in the diagnostic economy, while on the identical income paths the rational economy’s hazard is 1.5 percent. The mechanics are exactly Propositions 1 and 3 in sequence: the boom priced debt into the fragility corridor, and the first negative realization swings the belief tilt from extrapolated improvement to extrapolated decline, collapsing the price schedule and expanding the default set while debt is at its peak. Defaults follow booms in this model not because booms cause bad fundamentals but because booms cause fragile pricing.

6.5 Recalibration at the survey value of θ

Adding overreaction to a calibration built without it raised the default frequency to 11.6 per century, so the comparative statics leave one question standing: can the model fit the data with the belief distortion in place? I re-select the two parameters that Arellano (2008) chose to hit default moments, the discount factor and the default cost, by grid search at $\theta = 0.5$, targeting a default frequency of 3 per century and debt service of 5.5 percent of output. The search selects $\beta = 0.969$ and $\phi = 0.91$: a more patient government and a harsher default cost. Both movements are the direction the mechanism dictates. Diagnostic pricing makes booms cheap to borrow into, so the discipline that restores observed default frequencies must come from the government’s own patience and from the cost of walking away.

At the recalibrated parameters the economy defaults 3.1 times per century with mean debt service of 5.5 percent of output: both targets are hit. The mechanism survives in more moderate magnitudes. The within-cell news coefficient is -84 basis points per standard deviation against an exact zero for the rational economy solved at the identical parameters, and the reversal-quarter default hazard is 3.1 percent against 0.3 percent, a gap of the same order as in the unadjusted

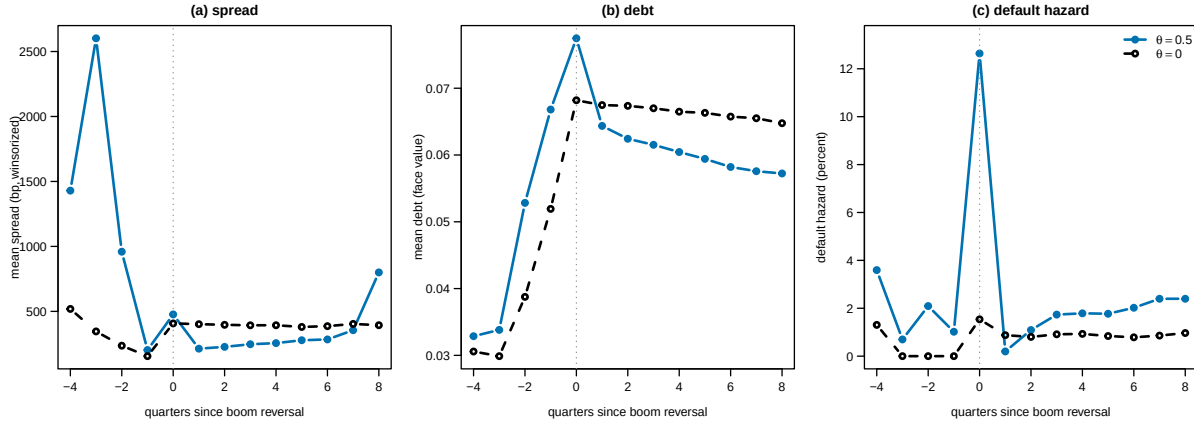


Figure 3: Boom-reversal episodes: at least three consecutive rises in log income followed by a fall, identified on the common simulated income path (5,593 episodes). Quarter 0 is the reversal. Spread and debt paths condition on continuous market access through the window. The default hazard does not.

economy. One number deserves flagging rather than burying: the rational economy at the recalibrated parameters defaults only 1 time per century. At parameters consistent with the observed default frequency under diagnostic pricing, the belief distortion accounts for roughly two thirds of defaults. Overreaction is not an ornament on a rational default model here, it is the model’s main engine of default.

Two limitations of the exercise are visible in the same numbers. The recalibrated economy’s spread levels are compressed (median 0.1 percent): this is the familiar failure of one-period-bond models, pushed further by the harsher default cost, and a full simulated-method-of-moments estimation with long-duration debt (Appendix E.2) is where spread levels should be fought for. And the search is a coarse grid over two parameters, not an estimation with standard errors. I report it as the disciplined minimum: the moments the mechanism was accused of breaking can be restored without breaking the mechanism.

Whether actual sovereign spreads behave this way is the next question.

7 Evidence from Spreads and Forecasts

The model makes four predictions that rational default pricing does not, each testable in a panel of emerging-market spreads matched to professional forecasts. Two are decisive, and I organize the section around that hierarchy. P1 is decisive because its rational null is an exact zero within cells, not a sign restriction. P4 is decisive because nothing in the construction of either panel forces the forecast estimates and the spread estimates to agree. P2 and P3 are supporting evidence: consistent with the mechanism, but with rational alternatives that are harder to exclude. The designs and the model’s predicted magnitudes follow, with results flagged as pending where they require the proprietary panels (J.P. Morgan EMBI spreads and Consensus Economics individual forecasts). Publishing the designs before the estimates is a feature of a first draft, not a defect. The rejection criteria are explicit: a within-cell news coefficient of zero at

conventional precision, or country-level news coefficients unrelated to country-level overreaction estimates, rejects the theory. Table 6 in Appendix D audits every dataset the designs require: coverage, frequency, access restrictions, and what can be shared in replication files.

P1 (decisive): news moves spreads conditional on fundamentals. The estimating equation, at country i and month t ,

$$s_{i,t} = \alpha_{c(i,t)} + \tau_t + \delta \text{Rev}_{i,t} + \Gamma' Z_{i,t} + e_{i,t}, \quad (14)$$

regresses the EMBI stripped spread on the consensus forecast revision of twelve-month-ahead real growth, where $\alpha_{c(i,t)}$ denotes fixed effects for *cells* of debt-to-GDP and output relative to trend (country-specific bins), τ_t absorbs global conditions including the VIX and United States rates, and Z holds the forecast *level*, ratings, a country-specific commodity terms-of-trade index, and the revision of the consensus fiscal-balance forecast, so that income news is not standing in for commodity news or fiscal news. Two features come straight from the theory. The cell fixed effects implement the nonparametric conditioning that Table 2 shows is necessary: with linear controls the rational null is not zero and the test is uninterpretable. And the forecast level belongs in Z because any rational model in which spreads respond to expected future income, such as Aguiar and Gopinath (2006) with trend shocks, makes the level informative. It is the *revision*, conditional on the level, that only overreaction prices. The model’s magnitude at $\theta = 0.5$ is 462 basis points per standard deviation of news at the comparative-static parameters and 84 at the recalibrated ones, and the rational prediction is zero at both. Sticky-information models predict the opposite sign for consensus data, which makes the test three-cornered rather than one-sided.

P2 (supporting): the effect is asymmetric and strongest near the boundary. Within the model, splitting ν into positive and negative parts gives within-cell responses of -657 and -368 basis points per standard deviation respectively, and interacting news with debt raises the response where debt is high. In the data this is (14) with Rev^+ , Rev^- , and $\text{Rev} \times \text{Debt}$ terms. I note one caveat in the model-to-data mapping: winsorization mutes the crash tail in the model statistics, so the model’s pooled asymmetry is a conservative benchmark.

P3 (supporting): returns reverse. Proposition 2 predicts that EMBI excess returns are low following positive revisions and high following negative ones, conditional on spreads themselves: the sovereign analogue of Greenwood and Hanson (2013). The regression of cumulative twelve-month excess returns on current revision histories, with the spread level controlled, has the rational null of zero once more. Risk-based stories require the *price* of risk to fall exactly when sentiment measures say optimism reigns, which is testable by adding conventional risk-premium proxies.

P4 (decisive): the belief parameter travels. Estimating θ_i country by country from forecasts (Section 5.2) and the news coefficient δ_i country by country from (14) gives a cross-sectional restriction: δ_i should be increasing in θ_i , and countries whose forecasters show no overreaction

Table 3: Empirical designs and status.

Test	Rational null	Model at $\theta = 0.5$	Status
P1: within-cell news coefficient	0	-462 bp per 1σ	pending data
P2: asymmetry, interaction with debt	0, 0	-657 / -368 bp	pending data
P3: return predictability from news	0	negative booms after	pending data
P4: δ_i increasing in θ_i	no relation	positive relation	pending data
CG regression, individual forecasts	0	-0.30	published for US ^a

Notes: ^aBordalo et al. (2020) report negative individual-level forecast-error-on-revision coefficients for most United States macroeconomic series. The emerging-market estimation is specified in Section 5.2. Data requirements: J.P. Morgan EMBI Global stripped spreads (Bloomberg or Datastream), Consensus Economics individual forecaster panel, IMF WEO vintages, CBOE VIX, sovereign ratings. Table 6 audits access and replication status.

should show no news effect. This is the falsification test I would most want to fail before believing the theory, because nothing in the fitting of either panel forces the two estimates to line up.

Table 3 collects the designs. Suppose they pass, and news moves sovereign spreads the way diagnostic pricing says. What, if anything, should a government do about it?

8 Fiscal Rules under Diagnostic Pricing

The instrument I study is the simplest one in the policy debate: a ceiling $\bar{b}$ on debt issuance, imposed permanently and known to all parties. For each ceiling on a grid I recompute the full equilibrium, price schedule included, and evaluate welfare as the value at zero debt and median income *under the true income process*, converting differences to basis points of permanent consumption. Because Section 2 taught that the answer turns on whose beliefs are distorted, I compute the complete two-by-two: government rational or diagnostic, market rational or diagnostic, with $\theta = 0.5$ wherever beliefs are diagnostic. A diagnostic government maximizes under its own tilted kernel (the Bellman equations are in Appendix E), and its welfare is then evaluated under the truth. Everything in this section is a numerical property of computed equilibria at the baseline calibration, conditional on that welfare criterion. Figure 4 plots the welfare gain of each ceiling in each cell, and Table 4 reports the best ceiling per cell.

The sign pattern is the result. When the government is rational, every ceiling hurts, and it hurts *more* when the market is diagnostic (-3.3 basis points at the tightest ceiling, against -2.3 in the fully rational cell): the rational government was selling overpriced debt to optimists, and the rule takes the trade away. When the government is diagnostic, ceilings help. Facing a diagnostic market the best ceiling yields +0.8 basis points and cuts the default frequency from 6.6 to 5.7 per century. Facing a rational market it yields +4.6. The gains are largest in the cell where the market is rational because there the government's extrapolation earns no offsetting rents:

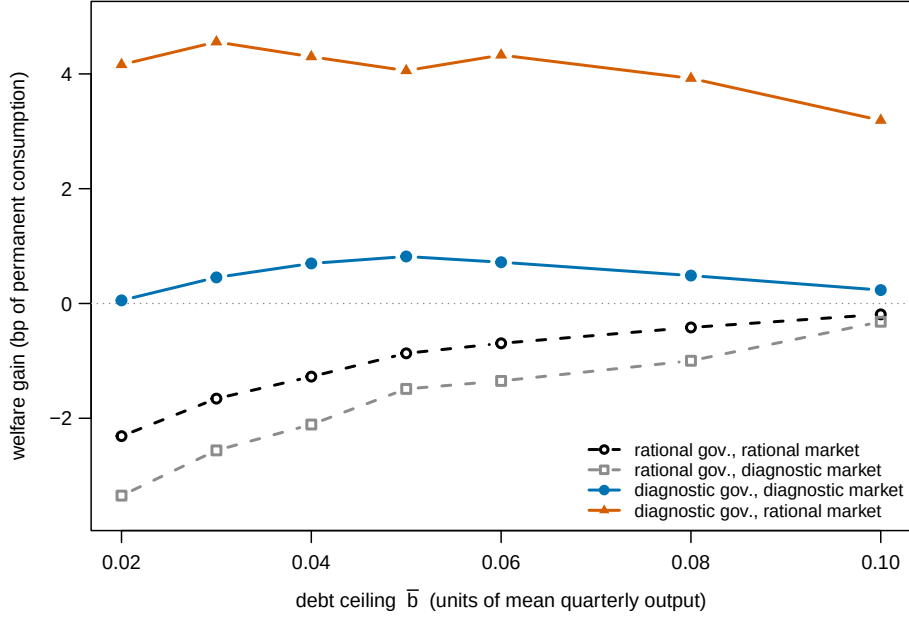


Figure 4: Welfare effect of a permanent debt ceiling $\bar{b}$, by configuration of beliefs, evaluated under the true income process at zero debt and median income. $\theta = 0.5$ where diagnostic.

Table 4: The whose-beliefs two-by-two: welfare gain of the best debt ceiling.

	Market rational	Market diagnostic
Government rational	-0.2 bp	-0.3 bp
Government diagnostic	+4.6 bp	+0.8 bp

Notes: basis points of permanent consumption, at the best ceiling on the grid $\{0.02, \dots, 0.10\}$ per cell. Negative entries mean every ceiling on the grid lowers welfare. $\theta = 0.5$ in diagnostic cells.

rational lenders price the optimistic government’s future behavior correctly, so its overborrowing (mean debt service 6.1 percent of output, against 4.4 in the rational benchmark) is pure loss, and the rule claws much of it back. Note also the diagnostic-government economies confirm the boom-overborrowing channel the two-period example could not deliver: with persistent income, a government that extrapolates good news borrows against an expected future that will not arrive, which is precisely the borrowing a good-news-contingent constraint would target.

The magnitudes are honest and small, single basis points of permanent consumption, for the usual reason: in this class of models default’s resource cost is modest and consumption smoothing is cheap. Two amplifications are foreseeable. Long-duration debt adds the dilution externality, which interacts with optimism because booms are exactly when dilution is cheapest (Appendix E sets up the recursion). And richer default costs of the Mendoza and Yue (2012) type scale every welfare number. The lesson itself should carry its conditions on its face: *under one-period debt, a benevolent government, and the computed equilibria at $\theta = 0.5$, a debt ceiling*

lowers sovereign welfare when the government prices the future correctly and raises it when the government extrapolates. In this environment fiscal rules are a commitment device against the government’s own extrapolation, not a correction of market mispricing. The mechanism behind the signs, rent extraction from optimistic lenders, is the analytical content of the two-period example. The welfare magnitudes are numerical properties of the baseline calibration, I have not recomputed the two-by-two at the recalibrated parameters, and whether the ranking survives long-duration debt is an open quantitative question. A corollary for institutional design: the state variable the model identifies, the run of recent news, is observable in real time from forecast revisions, so a rule that tightens after sustained upward revisions dominates a flat ceiling in principle. I leave the optimal state-contingent rule to later work. The flat ceiling suffices to establish the signs in the computed equilibria.

9 Conclusion

I have taken the workhorse model of sovereign default and replaced one primitive: creditors who overreact to news in the disciplined sense of diagnostic expectations, with the degree of overreaction measured from forecast surveys rather than fitted to prices. The cost was one observable state variable. The return was a set of predictions the rational model rules out exactly. Prices depend on the direction of recent news conditional on fundamentals, and bond returns are predictably poor after good news. Default regions expand when optimism reverses, so defaults cluster after booms on income paths where rational pricing generates almost none, and spread volatility rises severalfold alongside calmer medians. The predictions survive re-selecting the non-belief parameters to restore the observed default frequency, and at that data-consistent calibration the belief distortion accounts for roughly two thirds of defaults. Each prediction comes with an estimating equation and a pre-stated magnitude. The panels that would confirm or kill them are named, and the theory is written so that a zero within-cell news coefficient would falsify it.

The normative lesson runs against a common intuition. It is tempting to argue that if markets misprice sovereign risk, governments should be insulated from the market’s moods by fiscal rules. The model says nearly the reverse: a rational government profits from the market’s moods, and rules bind usefully only when the government shares them. Since the same surveys that discipline θ show official and professional forecasters overreacting, that condition is likely met in practice, but the argument for rules is then about the state’s own psychology, and the debate should be had on that ground.

The model’s scope should be stated plainly. The government here is a benevolent planner: the political economy of sovereign borrowing is outside the model, and the normative statements are conditional on that abstraction. Lenders are risk neutral, so every movement in spreads is probability rather than premium, and separating the two in the data is exactly the identification problem the return test confronts. Debt lasts one quarter. None of these abstractions is innocent, and the welfare magnitudes of Section 8 in particular should be read as displaying a mechanism,

not pricing a constitution.

Three extensions strike me as the productive ones. Long-duration bonds, where diagnostic booms should make dilution cheap exactly when optimism peaks, and where the interaction of the two distortions is an open quantitative question. Risk-averse lenders, where the identification problem of separating time-varying risk premia from belief distortions becomes serious and the forecast data must do the separating. And the estimation this draft specifies but does not yet contain: θ , country by country, from the forecasters who price these bonds. The model stands or falls with that number, which is where a theory of beliefs ought to stand or fall.

A Proofs

A.1 Proof of Lemma 1

Write $m_x = \rho x$ and $m_h = \rho h$. Up to factors that do not involve x' ,

$$\hat{p}_\theta(x' | x, h) \propto \exp \left\{ -\frac{(1+\theta)(x' - m_x)^2 - \theta(x' - m_h)^2}{2\sigma_\varepsilon^2} \right\}.$$

The coefficient on x'^2 in the numerator of the exponent is $(1+\theta) - \theta = 1$, so the variance is σ_ε^2 . The coefficient on $-2x'$ is $(1+\theta)m_x - \theta m_h$, so the mean is

$$(1+\theta)m_x - \theta m_h = \rho x + \theta \rho (x - h),$$

and the density, being Gaussian in x' with these two moments, is as claimed. ■

A.2 Proof of Lemma 2

By Lemma 1, $F_\theta(\cdot | x, h_i)$ is the Gaussian cdf with variance σ_ε^2 and mean $\mu_i = \rho x + \theta \rho \nu_i$. For $\nu_2 > \nu_1$ and $\theta > 0$, $\mu_2 > \mu_1$, and a Gaussian family with common variance is strictly ordered by first-order stochastic dominance in its mean: $F(z - \mu)$ is decreasing in μ for every z . ■

A.3 Proof of Lemma 3

Fix (x, h) and $b_2 > b_1$ in $\mathcal{B}$. For any issuance b' , consumption at b_2 is $y + q(b', x, h)b' - b_2$, strictly below its value at b_1 , and the continuation term in (8) does not involve inherited debt. With u strictly increasing, every candidate value is weakly lower at b_2 , strictly wherever consumption remains feasible, and the supremum over b' inherits the ordering: $V^R(b_2, x, h) \leq V^R(b_1, x, h)$, strictly whenever $V^R(b_1, x, h)$ is finite. V^D in (9) does not involve b . Hence $d = \mathbf{1}\{V^D > V^R\}$ is nondecreasing in b , and the default set is the upper interval $\{b \geq \underline{b}(x, h)\}$ with $\underline{b}(x, h)$ the smallest $b \in \mathcal{B}$ at which $V^D(x) > V^R(b, x, h)$. ■

A.4 Proof of Proposition 1 and Corollary 1

Fix (b', x) and write $g(x') = 1 - d(b', x', x)$, the repayment indicator next period. Under Assumption 1, g is nondecreasing. From (10), $Rq = \int g dF_\theta(\cdot | x, h)$. If $\nu_2 > \nu_1$, Lemma 2 gives first-order dominance, and the expectation of a nondecreasing function is nondecreasing under first-order dominance. Hence q is nondecreasing in ν .

For strictness, suppose $\theta > 0$ and default risk is interior at (b', x) : $0 < \int g dF_\theta < 1$. Because income is Markov on a totally ordered set and g is a nondecreasing indicator, interiority means g switches from 0 to 1 at some finite threshold $\underline{x}'(b', x)$ in the interior of the support. Writing $Z \sim N(0, \sigma_\varepsilon^2)$,

$$Rq(\nu) = \mathbb{E}[g(\mu(\nu) + Z)] = \mathbb{P}\{Z \geq \underline{x}' - \mu(\nu)\},$$

with $\mu(\nu) = \rho x + \theta \rho \nu$ strictly increasing in ν . Since the Gaussian distribution assigns positive density everywhere, the probability is strictly increasing in μ . This proves the proposition.

For the corollary: at $\theta = 0$ the kernel $\hat{p}_0(\cdot \mid x, h) = p(\cdot \mid x)$ does not involve h , so q is a function of (b', x) alone and any statistic of q computed within a (b', x) cell has zero variation in ν . Conversely, if $\theta > 0$ and default risk is interior at some (b', x) visited with positive probability, the strict monotonicity just established makes the within-cell slope strictly negative for spreads (spreads are strictly decreasing transformations of q). ■

A.5 Proof of Proposition 2

A lender pays $q(b', x, h)$ and receives $g(x') = 1 - d(b', x', x)$. The true-measure expected gross return is $\mathbb{E}_p[g \mid x]/q = R \mathbb{E}_p[g]/\mathbb{E}_{\hat{p}}[g]$, so

$$R^e(b', x, h) = R \frac{\mathbb{E}_p[g] - \mathbb{E}_{\hat{p}}[g]}{\mathbb{E}_{\hat{p}}[g]}.$$

At $\theta = 0$, $\hat{p} = p$ and $R^e = 0$ identically. For $\theta > 0$ and $\nu > 0$, $\hat{p}$ dominates p in first order (Lemma 2, comparing means $\rho x + \theta \rho \nu > \rho x$), so $\mathbb{E}_{\hat{p}}[g] \geq \mathbb{E}_p[g]$ and $R^e \leq 0$. The inequality reverses for $\nu < 0$. Monotonicity in ν follows because the numerator is nonincreasing and the denominator positive and nondecreasing in ν , both strictly where risk is interior by the argument in the previous proof. ■

A.6 Proof of Proposition 3

Take the equilibrium price schedule q and continuation value as given, and compare two histories $\nu_2 > \nu_1$ at the same (b, x) . For every candidate issuance $b' \geq 0$, consumption $y + q(b', x, h)b' - b$ is nondecreasing in ν by Proposition 1, and the continuation $\mathbb{E}[V(b', x', x) \mid x]$ does not involve h because the government's expectation is rational and tomorrow's reference is today's x . Hence each candidate's objective in (8) is nondecreasing in ν , and V^R , a pointwise supremum of nondecreasing functions of ν , is nondecreasing in ν . V^D in (9) involves neither b nor h . Therefore $d = \mathbf{1}\{V^D > V^R\}$ is nonincreasing in ν , which is the set inclusion claimed. ■

A.7 Proof of Proposition 4

Under the fixed-target convention, the diagnostic forecast of x_{t+1} made at t is $F_t x_{t+1} = \mathbb{E}_t x_{t+1} + \theta(\mathbb{E}_t x_{t+1} - \mathbb{E}_{t-1} x_{t+1})$ with $\mathbb{E}_t x_{t+1} = \rho x_t$ and $\mathbb{E}_{t-1} x_{t+1} = \rho^2 x_{t-1}$, so

$$F_t x_{t+1} = \rho x_t + \theta \rho \varepsilon_t, \quad F_{t-1} x_{t+1} = \rho^2 x_{t-1} + \theta \rho^2 \varepsilon_{t-1}.$$

Hence the revision and the error are

$$\text{Rev}_t = F_t x_{t+1} - F_{t-1} x_{t+1} = \rho(1 + \theta)\varepsilon_t - \theta \rho^2 \varepsilon_{t-1}, \quad \text{FE}_{t+1} = x_{t+1} - F_t x_{t+1} = \varepsilon_{t+1} - \theta \rho \varepsilon_t.$$

The innovations are i.i.d., so

$$\text{Cov}(\text{FE}_{t+1}, \text{Rev}_t) = -\theta(1 + \theta)\rho^2\sigma_\varepsilon^2, \quad \text{Var}(\text{Rev}_t) = \rho^2\sigma_\varepsilon^2[(1 + \theta)^2 + \theta^2\rho^2],$$

and the ratio is (12). The same algebra at horizon k (forecasting x_{t+k} , revising between $t - 1$ and t) gives $\text{Rev} = \rho^k(1 + \theta)\varepsilon_t - \theta\rho^{k+1}\varepsilon_{t-1}$ and $\text{FE} = \sum_{j=1}^k \rho^{k-j}\varepsilon_{t+j} - \theta\rho^k\varepsilon_t$, hence the *same* coefficient: the mapping is horizon invariant under this convention, which is convenient for fixed-event survey data. ■

B The Two-Period Example: Details

Consumption at date 0 is $c_0 = y_0 + qb$ with $q = \hat{\pi}/R$ in the risky region $\psi y_L < b \leq \psi y_H$. The objective is $\ln c_0 + \beta[\pi \ln(y_H - b) + (1 - \pi) \ln((1 - \psi)y_L)]$, and only the first two terms involve b . The first-order condition is

$$\frac{\hat{\pi}/R}{y_0 + (\hat{\pi}/R)b} = \frac{\beta\pi}{y_H - b},$$

and solving the linear equation gives (4). Differentiating (4) in $\hat{\pi}$ gives $\partial b^*/\partial \hat{\pi} = \beta\pi R y_0 / [\hat{\pi}^2(1 + \beta\pi)] > 0$. The second-order condition holds by concavity in b .

The safe–risky margin. The best safe plan solves $\max_{b \leq \psi y_L} \ln(y_0 + b/R) + \beta[\pi \ln(y_H - b) + (1 - \pi) \ln(y_L - b)]$ and its value W^S does not involve $\hat{\pi}$. The best risky plan's value $W^R(\hat{\pi})$ is strictly increasing in $\hat{\pi}$ by the envelope theorem, since $\partial W^R/\partial \hat{\pi} = u'(c_0) b^*/R > 0$. If parameters are such that $W^R(\pi) < W^S < \lim_{\hat{\pi} \rightarrow 1} W^R(\hat{\pi})$, there is a threshold $\hat{\pi}^\dagger$ at which the sovereign switches from safe to risky borrowing, and optimism beyond the threshold raises the true default probability from 0 to $1 - \pi$ discontinuously.

A diagnostic government in the example. If the government also prices the future with $\hat{\pi}$, the first-order condition becomes $(\hat{\pi}/R)/c_0 = \beta\hat{\pi} u'(y_H - b)$, in which $\hat{\pi}$ cancels from the price side and the belief side:

$$\hat{b} = \frac{y_H - \beta R y_0}{1 + \beta\hat{\pi}},$$

which is *decreasing* in $\hat{\pi}$. Probability optimism makes repayment subjectively more likely and so subjectively more expensive, and with log utility this cost effect outweighs the price effect. The example thus cannot be trusted for the sign of the diagnostic government's borrowing response: it contains only probability optimism. In the infinite-horizon model, optimism is about the *level* of future income, and the consumption-smoothing motive of an optimist is to borrow against the good future it expects. Quantitatively that channel dominates, and mean debt in the diagnostic-government cells exceeds the rational benchmark (Section 8). The contrast is a useful caution about two-period intuition in this class of models.

C Computation

Discretization. Log income uses a Tauchen (1986) chain with $n_x = 21$ states spanning ± 3 unconditional standard deviations. The diagnostic kernel is discretized by the same binning rule applied to the $N(\rho x_i + \theta \rho(x_i - x_k), \sigma_\varepsilon^2)$ density for every pair (i, k) , which is exact up to the binning because Lemma 1 keeps the tilted density Gaussian with unchanged variance. Debt lives on an evenly spaced grid of $n_b = 200$ points on $[0, 0.35]$ in units of mean quarterly output.

Algorithm. Given a price array q , one backward-induction sweep updates V^D , V^R (by full maximization over the debt grid), V , and d . The price is then updated from (10) and damped, $q \leftarrow \frac{1}{2}q + \frac{1}{2}q^{\text{new}}$. The joint iteration runs to sup-norm tolerances of 5×10^{-8} on both V and q , reached in 351 iterations at every θ , and a full solve takes roughly twenty seconds in multithreaded Julia on a laptop. At $\theta = 0$ the computed equilibrium reproduces the Arellano (2008) benchmark by construction.

Simulation. 500,000 quarters after a 1,000-quarter burn-in, with the income path drawn once and reused across every configuration (common random numbers), so that cross- θ comparisons and the event study condition on identical fundamentals. Statistics condition on market access. Spreads are annualized from quarterly prices and winsorized at the 1st and 99th percentiles for means and standard deviations.

Verification of Assumption 1 and grid robustness. In the computed equilibria at $\theta \in \{0, 0.5\}$ there is not a single violation of income-monotonicity, nor of the debt-monotonicity that Lemma 3 guarantees, across all grid combinations. At $\theta = 1$ a small fraction of grid columns shows a violation at extreme states with near-degenerate pricing: 0.02 percent at the baseline grids and 0.1 percent at doubled grids. The verification is not confined to the baseline environment. Doubling both grids ($n_x = 31$, $n_b = 400$) leaves zero violations at $\theta \in \{0, 0.5\}$. Replacing the asymmetric default cost with a proportional cost $y^D = (1 - \psi)y$ for $\psi \in \{0.02, 0.05\}$ leaves zero violations. All thirty parameter vectors visited by the recalibration search, and the recalibrated final pair, satisfy both monotonicities at every grid point. The headline moments are also stable under grid doubling: at $\theta = 0.5$ the within-cell news coefficient moves from -462 to -400 basis points per standard deviation, the default frequency from 11.6 to 11.4 per century, and the reversal-quarter hazard from 12.6 to 12 percent (1.5 to 1.1 for the rational economy), while the rational news coefficient remains zero (0). One statistic shows visible grid sensitivity: the winsorized spread standard deviation (57.8 against 35.3 percent), which inherits the extreme spread tail. I flag it rather than lean on it. The volatility ratio to the rational economy remains far above one at both densities, and no conclusion in the paper rests on its precise value.

Recalibration search. The grid behind Section 6.5 spans $\beta \in \{0.945, 0.953, 0.961, 0.969, 0.977\}$ and $\phi \in \{0.880, 0.895, 0.910, 0.925, 0.945, 0.969\}$, thirty pairs, each solved at $\theta = 0.5$ and simulated for 200,000 quarters. The loss function weights the squared relative miss of the default-frequency

Table 5: Calibration audit.

Parameter	Units, frequency	Category	Source or target (sample)	Range examined	Effect on headline results
$\gamma = 2$, eq. (8)	dimensionless	externally fixed	standard value in the quantitative default literature	not varied	scales welfare units, not signs
$\beta = 0.953$, eq. (8)	quarterly	internally calibrated	default moments at $\theta = 0$ (Arellano, 2008), re-selected at $\theta = 0.5$: 0.969	0.945–0.977	default frequency falls steeply in β (14.2 to 1.0 per century across the grid)
$r^* = 0.017$, eq. (10)	quarterly	externally fixed	U.S. five-year yield (Arellano, 2008)	not varied	shifts spread levels one for one
$\rho = 0.945$, eq. (5)	quarterly	directly estimated	detrended Argentine real GDP (Arellano, 2008)	mapping check	$\beta_{CG}(\theta)$ mapping nearly flat in ρ , tilt size scales with $\theta\rho$
$\sigma_\varepsilon = 0.025$, eq. (5)	quarterly	directly estimated	detrended Argentine real GDP (Arellano, 2008)	not varied	scales news size, not signs
$\lambda = 0.282$, eq. (9)	quarterly	externally fixed	market re-access delays (Arellano, 2008)	not varied	shortens exclusion, lowers effective default cost
$\phi = 0.969$, eq. (9)	$\mathbb{E}[y]$	internally calibrated	spread moments at $\theta = 0$ (Arellano, 2008), re-selected at $\theta = 0.5$: 0.91	0.880–0.969	governs default frequency and debt capacity jointly with β
$\theta = 0.5$, eq. (6)	dimensionless	scenario published range	in individual CG regressions (Bordalo et al., 2020), EM estimate pending (Section 5.2)	$\{0, 0.25, 0.5, 1\}$	mechanism vanishes exactly at $\theta = 0$, market access collapses near $\theta = 1$ at fixed (β, ϕ)

Notes: categories follow the four-way scheme of Section 6. No target moment used in calibration is reused for validation: the validation moments are the within-cell news coefficient, the return-predictability pattern, and the boom-reversal hazard profile, none of which enters the calibration or the recalibration loss. The region where the mechanism disappears is stated explicitly: all history dependence vanishes at $\theta = 0$, continuously, and the environment stops resembling emerging-market data near $\theta = 1$ at fixed non-belief parameters.

target (3 per century) four to one against the debt-service target (5.5 percent of output). Across the grid the default frequency ranges from 1.0 to 14.2 per century and debt service from 0.8 to 17.1 percent, so the two targets pin the pair down sharply, and the selected (0.969, 0.91) is interior to the grid. Final statistics reported in the text use 500,000 quarters at the selected pair, for $\theta = 0.5$ and for the rational benchmark at identical parameters.

Welfare. Consumption equivalents between configurations use $\Delta = (W_1/W_0)^{1/(1-\gamma)} - 1$ evaluated at zero debt and median income under the true measure. For diagnostic-government cells W is computed by policy evaluation as in Appendix E.

D Data Appendix

Spreads. J.P. Morgan EMBI Global stripped spreads, by country, monthly averages of daily quotes, accessed through Bloomberg or Datastream. The stripped spread nets out collateralized

cash flows and is the standard market-risk measure in this literature (Uribe and Yue, 2006; Longstaff et al., 2011). Spreads are winsorized at the 1st and 99th percentiles within country. Default and restructuring windows are flagged using the Cruces and Trebesch (2013) episode list and excluded from pricing regressions, mirroring the model’s conditioning on market access.

Forecasts. Consensus Economics monthly surveys, individual-forecaster level, of real GDP growth for the current and next calendar year, available for the major emerging markets from the early-to-mid 1990s. Fixed-event forecasts are converted to a fixed twelve-month horizon by weighting the two events by months remaining in the year: $F_t^{12m} = \frac{m}{12}F_t^y + (1 - \frac{m}{12})F_t^{y+1}$, with m the months left in year y . Revisions are month-over-month changes in F^{12m} , and the fixed-target revisions used for the Coibion–Gorodnichenko regressions are computed within event year. Realized growth is the *first-release* outturn from the IMF World Economic Outlook vintage database, so that forecast errors are not contaminated by later data revisions.

Fundamentals and controls. General government debt to GDP comes from the WEO, and output relative to trend from a country-specific quadratic trend in log real GDP (robustness: Hamilton filter). The CBOE VIX and the ten-year United States Treasury yield serve as global controls, absorbed by time effects in the baseline. Standard & Poor’s long-term foreign-currency ratings are mapped to a 21-notch scale. One timing mismatch deserves notice: WEO debt is annual, and interpolating it to monthly frequency puts measurement error inside the conditioning cells. The design therefore uses lagged debt for cell assignment to avoid look-ahead, and quarterly BIS debt data cover a subsample for robustness.

Sample. The intersection of EMBI Global membership and Consensus coverage with at least ten years of joint data, roughly 25 countries including Argentina, Brazil, Bulgaria, Chile, China, Colombia, Ecuador, Hungary, Indonesia, Malaysia, Mexico, Nigeria, Panama, Peru, the Philippines, Poland, Russia, South Africa, Turkey, Ukraine, and Venezuela, monthly, 1994–2024. Cell fixed effects for (14) use country-specific terciles of debt-to-GDP crossed with terciles of output relative to trend, with finer bins as a robustness dimension. Standard errors are clustered two-way by country and month, with Driscoll–Kraay errors as robustness given overlapping forecast horizons.

Table 6: Data feasibility audit.

Dataset	Coverage	Frequency	Access	Observed or proxied	Replication
EMBI Global stripped spreads (J.P. Morgan)	~25 countries, 1994–2024	daily, used monthly	Bloomberg or Datastream license	observed	code shareable, raw data licensed
Consensus Economics, individual forecasters	major emerging markets, mid-1990s–2024	monthly	commercial license	observed	code shareable, raw data licensed
IMF WEO vintage database	all sample countries	semiannual vintages	public	observed	shareable
WEO debt to GDP	all sample countries	annual	public	interpolated to monthly (proxy)	shareable
S&P foreign-currency ratings	all sample countries	event dates	licensed	mapped to 21-notch scale	derived series shareable
CBOE VIX, U.S. Treasury yields	global	daily	public	observed	shareable
Cruces and Trebesch (2013) restructurings	episodes, 1970–2013	episode	public	observed	shareable

Notes: the binding constraints are the two licensed panels. Both are standard in the literature and available at the institutions where this work would be conducted. Replication files can include all code, all public data, and exact extraction instructions for the licensed series.

E Extensions

E.1 The diagnostic government

When the government itself holds diagnostic beliefs, its expectations in (8)–(9) are taken under $\hat{p}_{\theta_g}(\cdot | x, h)$, and the default value acquires the belief state: the Bellman system becomes

$$V^R(b, x, h) = \max_{b'} \left\{ u(y + q(b', x, h)b' - b) + \beta \int V(b', x', x) \hat{p}_{\theta_g}(x' | x, h) dx' \right\}, \quad (\text{E.1})$$

$$V^D(x, h) = u(y^D(x)) + \beta \int [\lambda V(0, x', x) + (1 - \lambda)V^D(x', x)] \hat{p}_{\theta_g}(x' | x, h) dx', \quad (\text{E.2})$$

with lenders pricing under $\hat{p}_{\theta_\ell}$ as before. The formulation treats the government as sophisticated about its own future beliefs in the stationary sense: it correctly anticipates that its future self will evaluate continuation problems with the tilt implied by the future state (x', x) . Because the tilted kernel is Markov on (x, h) , the fixed point is well defined. The alternative, anticipated-utility formulation in which the government treats today’s tilted forecast as permanent gives similar quantitative answers in related settings and is a robustness item. *True* welfare of the induced policies (d, b') solves the linear system

$$W(b, x, h) = \begin{cases} W^D(x), & d(b, x, h) = 1, \\ u(c(b, x, h)) + \beta \int W(b'(b, x, h), x', x) p(x' | x) dx', & d = 0, \end{cases}$$

with W^D defined analogously under p , and all welfare statements in Section 8 use W .

E.2 Long-duration bonds

With the Chatterjee and Eyigungor (2012) perpetuity (coupon κ , principal decay δ), the lenders' valuation satisfies

$$q(b', x, h) = \frac{1}{R} \widehat{\mathbb{E}}_{\theta} \left[(1 - d(b', x', x)) (\kappa + (1 - \delta) q(b''(b', x', x), x', x)) \mid x, h \right],$$

where b'' is the successor government's issuance policy. Two diagnostic forces now compound. Today's lenders tilt the distribution of x' . But the resale price inside the expectation is itself evaluated at the future belief state (x', x) , so optimistic lenders are optimistic about future *sentiment* as well as future income. Booms therefore raise long-bond prices more than short-bond prices, and the dilution externality, the incumbent's incentive to issue against the value of legacy bonds, is cheapest to exercise exactly at peak optimism. I conjecture the interaction is quantitatively large. Computing it requires only adding the h dimension to the standard long-bond algorithm, and it is the natural second chapter of this agenda.

E.3 Reference-belief conventions

The model's reference density is $p(\cdot \mid h)$, the kernel evaluated at yesterday's state, which keeps the belief state two-dimensional and Lemma 1 exact. Bordalo et al. (2020) define the reference as the lagged forecast of the same target, giving the tilted mean $\rho x + \theta \rho(x - \rho h) = \rho x + \theta \rho \varepsilon_t$. The two means differ by $\theta \rho(1 - \rho)h$, which is $O(\theta(1 - \rho))$ and about five percent of the tilt at the calibrated persistence. Every result in Section 4 goes through under the alternative convention with ν redefined as the innovation ε , since the FOSD ordering in Lemma 2 needs only a mean tilt that is increasing in news. Proposition 4 is stated under the survey convention, so no adjustment is needed on the measurement side, and the horizon invariance shown in its proof covers the fixed-event structure of the Consensus data after the conversion of Appendix D.

E.4 Additional quantitative results

At $\theta = 1$ the economy defaults 29.3 times per century and holds market access only 73.9 percent of the time. Median spreads collapse to 0.5 percent because the sovereign borrows very little in normal times, while the winsorized standard deviation explodes. I read the $\theta = 1$ column as diagnostic of misspecification at fixed non-belief parameters rather than as a candidate description of the data, and it disciplines how far the survey estimates for other asset classes, such as the $\theta \approx 0.9$ of Bordalo et al. (2018) for credit cycles, can be imported into a sovereign default environment without re-estimation. The full ceiling grids behind Table 4 are in Figure 4. The optimal ceilings are $\bar{b} = 0.05$ (market diagnostic) and $\bar{b} = 0.03$ (market rational) in the diagnostic-government cells, roughly 1.2 and 0.7 percent of annual output. Replication code (Julia for the equilibrium computation, R for figures) accompanies the draft, and a full solve-and-simulate cycle for all configurations runs in under a minute.

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